

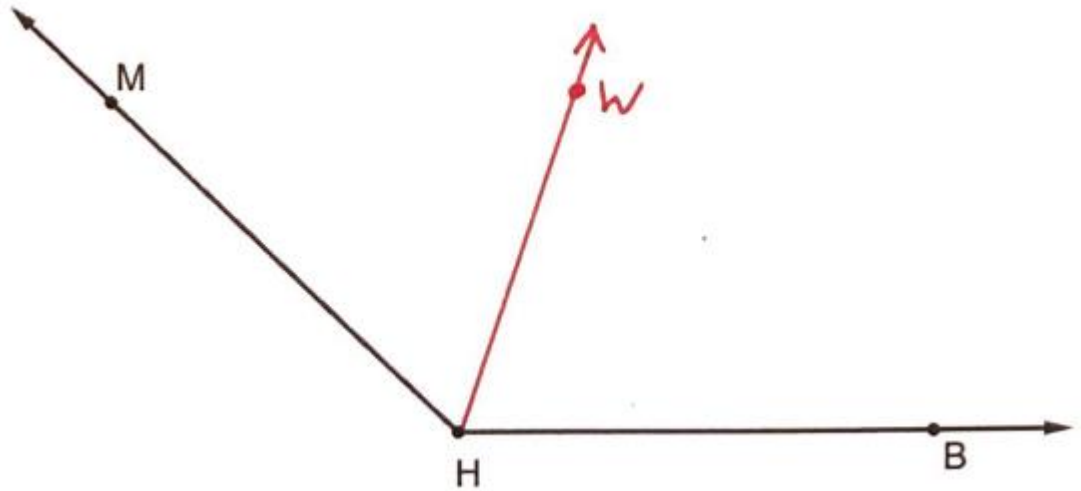
Geometry
Unit 1
Lesson 12 Practice

Name Answer Key
Date _____
Period _____

$\angle MHB$ is shown.

1. Fold the paper so that $\overrightarrow{HM}$ and $\overrightarrow{HB}$ lie on each other.

2. Use the fold to draw $\overrightarrow{HW}$.



3. Complete the statement.

$\overrightarrow{HW}$ bisects angle MHB so that $\angle$ MHW $\cong$ $\angle$ BHW .

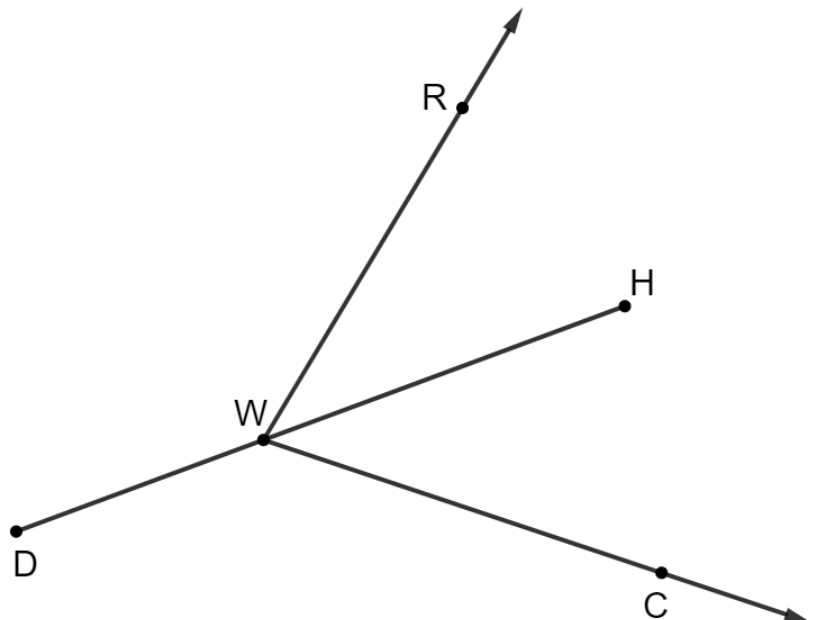
Line segment DH bisects $\angle RWC$ such that $m\angle RWH = (7k + 16.3)^\circ$ and $m\angle CWH = (13k - 2.9)^\circ$.

4. Determine the value of k .

$$\begin{aligned} 7k + 16.3 &= 13k - 2.9 \\ 16.3 &= 6k - 2.9 \\ 19.2 &= 6k \\ k &= 3.2 \end{aligned}$$

5. Determine $m\angle RWH$.

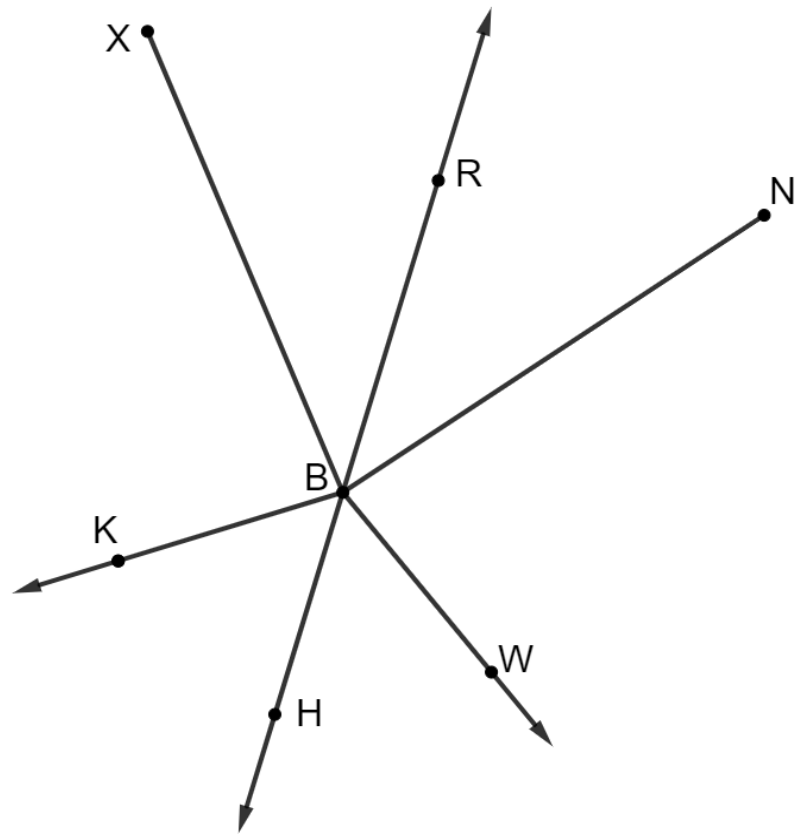
$$\begin{aligned} m\angle RWH &= 7(3.2) + 16.3 \\ &= 22.4 + 16.3 = 38.7^\circ \end{aligned}$$



6. Determine $m\angle CWD$.

$$m\angle CWD = 180 - 38.7 = 141.3^\circ$$

Line HR bisects $\angle XBN$ and $\angle KBW$ such that $m\angle XBR = (2x + 7.5)^\circ$, $m\angle KBH = (3y - 7)^\circ$, $m\angle WBH = (7y - 91)^\circ$, and $m\angle XBN = (5x - 1)^\circ$.



7. Determine the value of x .

$$2 \cdot m\angle XBR = m\angle XBN$$

$$2(2x + 7.5) = 5x - 1$$

$$4x + 15 = 5x - 1$$

$$15 = x - 1$$

$$16 = x$$

8. Determine the value of y .

$$3y - 7 = 7y - 91$$

$$-7 = 4y - 91$$

$$84 = 4y$$

$$21 = y$$

9. Determine $m\angle RBN$.

$$m\angle RBN = m\angle XBR = 2x + 7.5 = 2(16) + 7.5 = 32 + 7.5 = 39.5^\circ$$

10. Determine $m\angle KBW$.

$$m\angle KBW = 2(m\angle KBH) = 2(3 \cdot 21 - 7) = 2(63 - 7) = 2(56) = 112^\circ$$